

EX:	
DATE:	

AIM:

ALGORITHM:

CODE:

```
import pandas as pd  
df=pd.read_csv('Sleep_health_and_lifestyle_dataset.csv')  
print(df)
```

	Person ID	Gender	Age	Occupation	Sleep Duration \
0	1	Male	27	Software Engineer	6.1
1	2	Male	28	Doctor	6.2
2	3	Male	28	Doctor	6.2
3	4	Male	28	Sales Representative	5.9
4	5	Male	28	Sales Representative	5.9
..	...	...	...	...	...
369	370	Female	59	Nurse	8.1
370	371	Female	59	Nurse	8.0
371	372	Female	59	Nurse	8.1
372	373	Female	59	Nurse	8.1
373	374	Female	59	Nurse	8.1

	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category \
0	6	42	6	Overweight
1	6	60	8	Normal
2	6	60	8	Normal
3	4	30	8	Obese
4	4	30	8	Obese
..	...	...	...	...
369	9	75	3	Overweight
370	9	75	3	Overweight
371	9	75	3	Overweight
372	9	75	3	Overweight
373	9	75	3	Overweight

	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	126/83	77	4200	NaN
1	125/80	75	10000	NaN
2	125/80	75	10000	NaN
3	140/90	85	3000	Sleep Apnea
4	140/90	85	3000	Sleep Apnea
..	...	...	...	...
369	140/95	68	7000	Sleep Apnea
370	140/95	68	7000	Sleep Apnea
371	140/95	68	7000	Sleep Apnea
372	140/95	68	7000	Sleep Apnea
373	140/95	68	7000	Sleep Apnea

[374 rows x 13 columns]

```
print(df.head())
```

	Person ID	Gender	Age	Occupation	Sleep Duration	\
0	1	Male	27	Software Engineer	6.1	
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2	125/80	75	10000	NaN
3	140/90	85	3000	Sleep Apnea
4	140/90	85	3000	Sleep Apnea

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 374 entries, 0 to 373
Data columns (total 13 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Person ID                            374 non-null    int64
1   Gender                               374 non-null    object
2   Age                                  374 non-null    int64
3   Occupation                           374 non-null    object
4   Sleep Duration                       374 non-null    float64
5   Quality of Sleep                     374 non-null    int64
6   Physical Activity Level               374 non-null    int64
7   Stress Level                         374 non-null    int64
8   BMI Category                         374 non-null    object
9   Blood Pressure                       374 non-null    object
10  Heart Rate                           374 non-null    int64
11  Daily Steps                          374 non-null    int64
12  Sleep Disorder                       155 non-null    object
dtypes: float64(1), int64(7), object(5)
memory usage: 38.1+ KB
None
```

```
print(df.describe())
```

	Person ID	Age	Sleep Duration	Quality of Sleep \	
count	374.000000	374.000000	374.000000	374.000000	
mean	187.500000	42.184492	7.132086	7.312834	
std	108.108742	8.673133	0.795657	1.196956	
min	1.000000	27.000000	5.800000	4.000000	
25%	94.250000	35.250000	6.400000	6.000000	
50%	187.500000	43.000000	7.200000	7.000000	
75%	280.750000	50.000000	7.800000	8.000000	
max	374.000000	59.000000	8.500000	9.000000	

	Physical Activity Level	Stress Level	Heart Rate	Daily Steps
count	374.000000	374.000000	374.000000	374.000000
mean	59.171123	5.385027	70.165775	6816.844920
std	20.830804	1.774526	4.135676	1617.915679
min	30.000000	3.000000	65.000000	3000.000000
25%	45.000000	4.000000	68.000000	5600.000000
50%	60.000000	5.000000	70.000000	7000.000000
75%	75.000000	7.000000	72.000000	8000.000000
max	90.000000	8.000000	86.000000	10000.000000

```
!pip install openpyxl
```

```
Requirement already satisfied: openpyxl in C:\Users\DELL\AppData\Local\Programs\Python\Python313\Lib\site-packages (3.1.5)
Requirement already satisfied: et-xmlfile in C:\Users\DELL\AppData\Local\Programs\Python\Python313\Lib\site-packages (from openpyxl) (2.0.0)
```

```
df = pd.read_csv("Sleep_health_and_lifestyle_dataset.csv")
df.to_excel("Sleep_health_and_lifestyle_dataset.xlsx", index=False) print("Excel file created
successfully.")
```

```
Excel file created successfully.
```

```
df_excel = pd.read_excel("Sleep_health_and_lifestyle_dataset.xlsx")
print(df_excel.head())
```

	Person ID	Gender	Age	Occupation	Sleep Duration	\
0	1	Male	27	Software Engineer	6.1	
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4	140/90	85	3000	Sleep Apnea

```

import sqlite3

df = pd.read_csv("Sleep_health_and_lifestyle_dataset.csv")

conn = sqlite3.connect("sleep_health.db")

df.to_sql("sleep_data", conn, if_exists="replace", index=False)

print("Data imported into SQL successfully!")

query = "SELECT * FROM sleep_data"

df_sql = pd.read_sql(query, conn)

print(df_sql.head())

```

Data imported into SQL successfully!

	Person ID	Gender	Age	Occupation	Sleep Duration	\
0	1	Male	27	Software Engineer	6.1	
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	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
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2	125/80	75	10000	None
3	140/90	85	3000	Sleep Apnea
4	140/90	85	3000	Sleep Apnea

```
url="Sleep_health_and_lifestyle_dataset.csv"
```

```
df=pd.read_csv(url)
```

```
print(df.head())
```

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1	125/80	75	10000	NaN
2	125/80	75	10000	NaN
3	140/90	85	3000	Sleep Apnea
4	140/90	85	3000	Sleep Apnea

```
df=pd.read_csv("Sleep_health_and_lifestyle_dataset.csv")  
df.to_csv("Sleep_Output.csv", index=False)  
print("CSV File Exported Successfully")
```

CSV File Exported Successfully

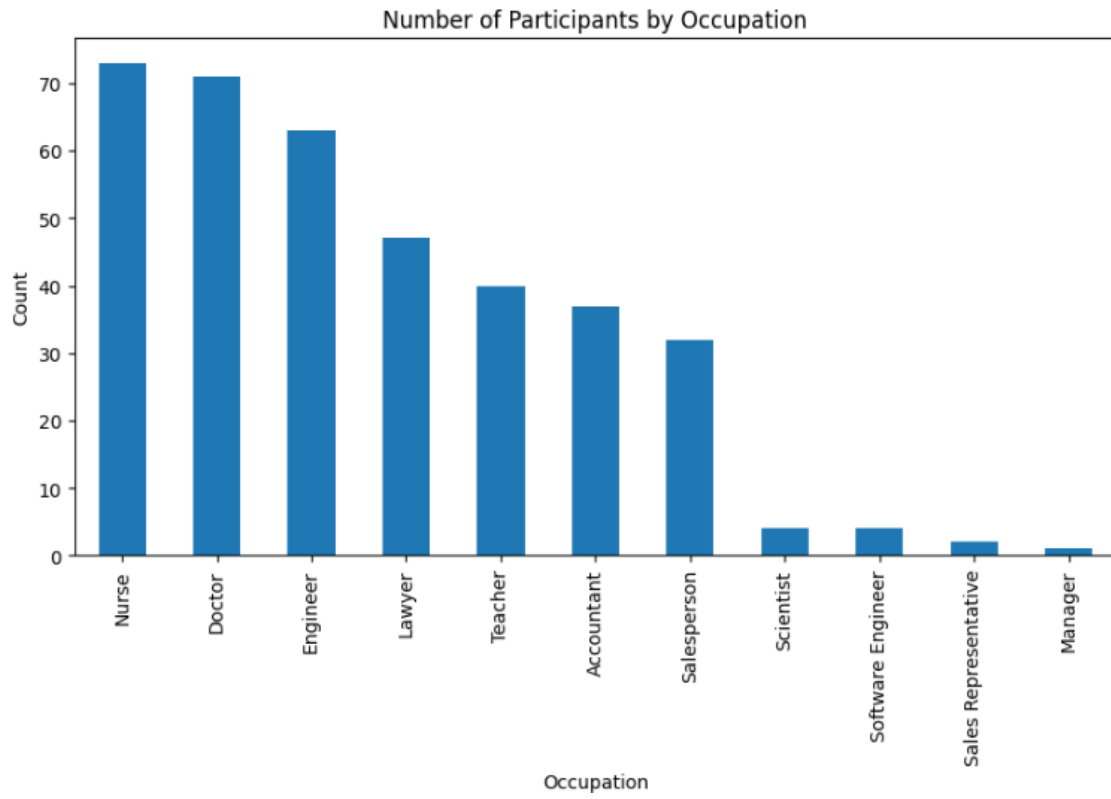
```
df=pd.read_csv("Sleep_health_and_lifestyle_dataset.csv")  
df.to_excel("Sleep_Output.xlsx", index=False)  
print("Excel File Exported")
```

Excel File Exported

```
df=pd.read_csv("Sleep_health_and_lifestyle_dataset.csv")  
df.to_json("Sleep_Output.json", orient="records")  
print("JSON File Exported")
```

JSON File Exported

```
import matplotlib.pyplot as plt  
  
df=pd.read_csv("Sleep_health_and_lifestyle_dataset.csv")  
  
occupation_count = df["Occupation"].value_counts()  
occupation_count.plot(kind="bar", figsize=(10,5))  
  
plt.title("Number of Participants by Occupation")  
  
plt.xlabel("Occupation")  
  
plt.ylabel("Count")  
  
plt.show()
```



RESULT: